Precalculus

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Chapter 1

Relations and Functions

1.1 Sets of Real Numbers and the Cartesian Coordinate Plane

1.1.1 Sets of Numbers

Definition 1.1.1 (Set)**.** A **set** is a well-defined collection of objects which are called the 'elements' of the set. Here, 'well-defined' means that it is possible to determine if something belongs to the collection or not, without prejudice.

Definition 1.1.2. Ways to describe sets

1. **The Verbal Method**: Use a sentence to define a set.
2. **The Roster Method**: Begin with a left brace '{', list each element of the set only once and then end with a right brace '}'.
3. **The Set-Builder Method**: A combination of the verbal and roster methods using a "dummy variable" such as x.

Definition 1.1.3. Sets of Numbers

1. The **Empty Set**: $\emptyset = \{\} = \{x \mid x \neq x\}$. This is the set with no elements.
2. The **Natural Numbers**: $\mathbb{N} = \{1, 2, 3, ...\}$. The periods of ellipsis here indicate that the natural numbers contain 1, 2, 3, 'and so forth'.
3. The **Whole Numbers**: $\mathbb{W} = \{0, 1, 2, ...\}$
4. The **Integers**: $\mathbb{Z} = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$
5. The **Rational Numbers**: $\mathbb{Q} = \{\frac{a}{b} \mid a \in \mathbb{Z} \wedge b \in \mathbb{Z}\}$. Rational numbers are the ratios of integers.
6. The **Real Numbers**: $\mathbb{R} = \{x \mid x \text{ possesses a decimal representation.}\}$

7. The **Irrational Numbers**: $\mathbb{P} = \{x \mid x \text{ is a non-rational real number.}\}$ Said another way, an irrational number is a decimal which neither repreats nor terminates.

8. The **Complex Numbers**: $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R} \wedge i = \sqrt{-1}\}$.

Definition 1.1.4 (Intervals)**.** Visualizing $\mathbb{R}$ as a line, segments on this line are called **intervals** of numbers. An interval can be described as a set in the form $\{x \mid a < x < b\}$, or in interval notation (a, b)

Definition 1.1.5 (Set Operations)**.** Supposing that A and B are two sets.

- The **intersection** of A and B: $A \cap B = \{x \mid x \in A \wedge x \in B\}$
- The **union** of A and B: $A \cup B = \{x \mid x \in A \vee x \in B\}$

Definition 1.1.6 (Cartesian Coordinate System)**.** The **Cartesian Coordinate System** is a coordinate system that specifies each point on it uniquely by a pair of real numbers called coordinates. In a Cartesian Coordinate Plane:

- (a, b) and (c, d) represent the same point in the plane if and only if $a = c \wedge b = d$
- (x, y) lies on the *x-axis* if and only if $y = 0$
- (x, y) lies on the *y-axis* if and only if $x = 0$
- The origin is the point $(0, 0)$. It is the only point common to both axes.

Definition 1.1.7 (Symmmetry)**.** Two points (a, b) and (c, d) in the plane are said to be:

- **symmetric about the *x-axis*** if $a = c \wedge b = -d$
- **symmetric about the *y-axis*** if $a = -c \wedge b = d$
- **symmetric about the *origin*** if $a = -c \wedge b = -d$

Definition 1.1.8 (Reflections)**.** To reflect a point (x, y) about the:

- *x-axis*, replace y with $-y$
- *y-axis*, replace x with $-x$
- *origin*, replace x with $-x$ and y with $-y$

Definition 1.1.9 (Distance)**.** Supposing we hace two points $P(x_0, y_0)$ and $Q(x_1, y_1)$ in the plane. The **distance** between P and Q is the length of the line segment joining P with Q. Using the pythagorean theorem we can derive the distance formula as:

$$d = \sqrt{(x_1 - x_0)^2 + (y_1 - y_0)^2}$$

Definition 1.1.10. The midpoint M of the line segment connecting $P(x_0, y_0)$ and $Q(x_1, y_1)$ is:

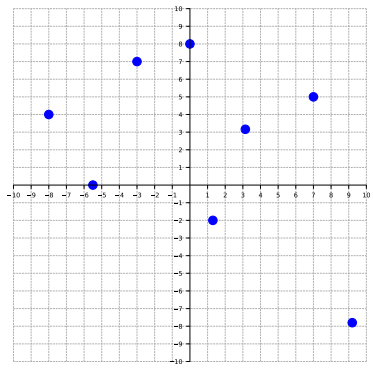
$$M = (\frac{x_0 + x_1}{2}, \frac{y_0 + y_1}{2})$$

1.1.2 Problems

2. $(-1, 5] \cap [0, 8) = [0, 5]$

8. $(-\infty, 5) \cup (5, \infty)$

20.



22.

$$d = \sqrt{(-3-1)^2 + (5-2)^2} = \sqrt{16+9} = \sqrt{25} = 5$$
$$M = (\frac{1-3}{2}, \frac{2+5}{2}) = (-1, 3.5)$$

30.

$$4 = \sqrt{(3-x)^2 + (2+1)^2} = \sqrt{9 - 6x + x^2 + 9} = \sqrt{x^2 - 6x + 18}$$
$$x^2 - 6x + 18 = 16$$
$$x^2 - 6x + 2 = 0$$
$$x_1 = 3 + \sqrt{7}, x_2 = 3 - \sqrt{7}$$
$$(3 + \sqrt{7}, -1), (3 - \sqrt{7}, -1)$$

34. He's at $(-3, -4)$, at a distance of 5 miles, after running he will be at $(4, -4)$

35. (a)
$$d = \sqrt{(a-a)^2 + (y_1 - y_0)^2} = \sqrt{(y_1 - y_0)^2} = y_1 - y_0$$

(b)
$$d = \sqrt{(x_1 - x_0)^2 + (b-b)^2} = \sqrt{(x_1 - x_0)^2} = x_1 - x_0$$

(c)
$$d = \sqrt{(x-x)^2 + (y-y)^2} = \sqrt{0} = 0$$

36.

$$M = (\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2})$$

$$\begin{aligned}
d_{PM} &= \sqrt{(\frac{x_1 + x_2}{2} - x_1)^2 + (\frac{y_1 + y_2}{2} - y_1)^2} \\
&= \sqrt{(\frac{x_2 - x_1}{2})^2 + (\frac{y_2 - y_1}{2})^2} \\
&= \sqrt{\frac{x_1^2 - 2x_1x_2 + x_2^2}{4} + \frac{y_1^2 - 2y_1y_2 + y_2^2}{4}} \\
d_{QM} &= \sqrt{(\frac{x_1 + x_2}{2} - x_2)^2 + (\frac{y_1 + y_2}{2} - y_2)^2} \\
&= \sqrt{(\frac{x_1 - x_2}{2})^2 + (\frac{y_1 - y_2}{2})^2} \\
&= \sqrt{\frac{x_1^2 - 2x_1x_2 + x_2^2}{4} + \frac{y_1^2 - 2y_1y_2 + y_2^2}{4}}
\end{aligned}$$

$$d_{PM} = d_{QM}$$

37. To prove that the points are the vertices of a right triangle there must be an internal angle with a value of 90deg or prove the converse of the pythagorean theorem.

    (a)

    $$AB = \sqrt{13}, AC = 2\sqrt{13}, BC = \sqrt{65}$$
    $$BC^2 = 65 = AB^2 + AC^2 = 13 + 52$$

    (b)

    $$AB = 5\sqrt{2}, AC = 5, BC = 5$$
    $$AB^2 = 50 = AC^2 + BC^2 = 25 + 25$$

38. To find the remaining point that makes the square, we must find a point that is at the same distance as the other points. We can create a vector from one of the sides of the square and just use it as a base to create the square. If we take $CB = (4, 3)$, then applying that to A, we get $D = (1, 4)$

1.2 Relations

Definition 1.2.1 (Relation)**.** A **relation** is a set of points in the plane.

Definition 1.2.2 (The Fundamental Graphing Principle)**.** Supposing there exists a relation R. Plotting the points of the set R in the plane, produces the **graph** of R

Definition 1.2.3 (The Fundamental Graphing Principle)**.** The graph of an equation is the set of points which satisfices the equation that describes the relation. That is, a point (x, y) is on the graph of the equation if and only if x and y satisfy the equation.

Definition 1.2.4 (Intercepts)**.** The places where a graph crosses or touches the axes are called **intercepts**, intercepts come in two distinct varieties:

- A point on a graph which is also on the *x-axis* is called an **x-intercept** of the graph
- A point on a graph which is also on the *y-axis* is called an **y-intercept** of the graph

Definition 1.2.5 (Finding the Intercepts of the Graph of an Equation)**.** Given an equation involving x and y, we find the intercepts of the graph as follows:

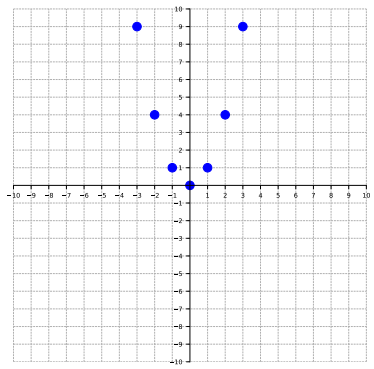
- *x-intercepts* have the form $(x, 0)$; set $y = 0$ in the equation and solve for x
- *y-intercepts* have the form $(0, y)$; set $x = 0$ in the equation and solve for y

Definition 1.2.6 (Testing the Graph of an Equation for Symmetry)**.** To test the graph of an equation for Symmetry

- about the *y-axis* - substitute $(-x, y)$ into the equation and simplify. If the result is equivalent to the original equation, the graph is symmetric about the *y-axis*.
- about the *x-axis* - substitute $(x, -y)$ into the equation and simplify. If the result is equivalent to the original equation, the graph is symmetric about the *x-axis*.
- about the *origin* - substitute $(-x, -y)$ into the equation and simplify. If the result is equivalent to the original equation, the graph is symmetric about the *origin*.

1.2.1 Problems

1.



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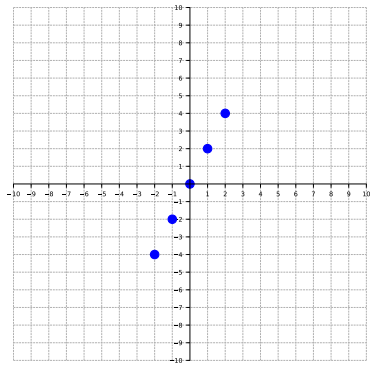

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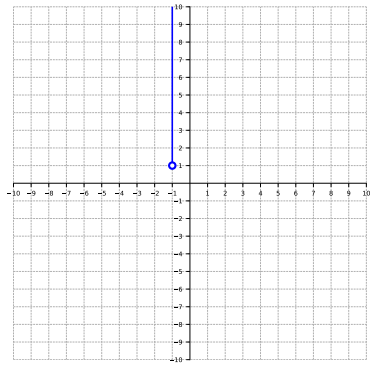


21.

$$\{(-4,-1),(-2,1),(0,3),(1,4)\} = \{(m, m+3) \mid m \in \{-4,-2,0,1\}\}$$

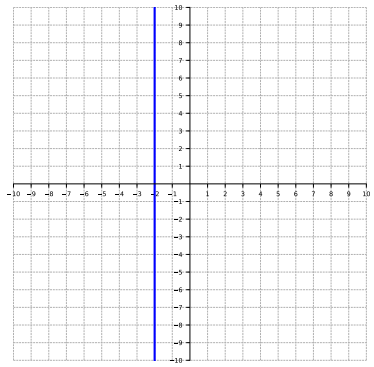
23.

$$\{(2,y) \mid y > -3\}$$

28.

$$\{(x,y) \mid -3 < x \leq 2\}$$

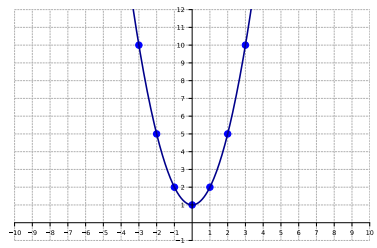
31.



37. We can start by plotting all the points for which y = 0, that is all the odd integers, then we just copy those points and translate them into the even integers on y. Will look like a grid pattern. Points in the relation: $(1,0),(3,2)$, Points not in the relation $(1,1),(2,2)$

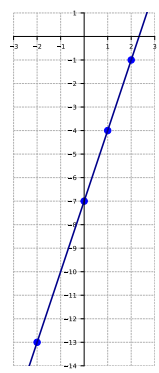
39. Practically the it is the graph of the equation $y = 2^x$

41.
- *x-intercept* doesn't exists, *y-intercept* at 0
-



- Graph is symmetric about the *y-axis*

47.
- *x-intercept* at $\frac{7}{3}$, *y-intercept* at -7
-



- Graph is not symmetric, example using $-x$ we see that $y = 3x - 7 \neq -3x - 7$

53. Multiple values of x can be multiple values of y and viceversa

1.3 Introduction to Functions

Definition 1.3.1 (Function)**.** A relation in which each *x-coordinate* is matched with only one *y-coordinate* is said to describe y as a **function** of x

Definition 1.3.2 (Vertical Line Test)**.** A set of points in the plane represents a y as a function of x if and only if no two points lie on the same vertical line.

Definition 1.3.3 (Domain and Range of a function)**.** Suppose that F is a relation which describes y as a function of x

- The set of the *x-coordinates* of the points in F is called the **domain** of F
- The set of the *y-coordinates* of the points in F is called the **range** of F

1.3.1 Problems

1. Function, domain: $\{-3, -2, ..., 2, 3\}$, range: $\{0, 1, 4, 9\}$

5. Not a function, example $(1, 2)$ and $(1, 4)$

11. Function, domain $(-\infty, \infty)$, range $[0, \infty)$

13. Function, domain $[-4, 1]$, range $[-1, 4]$

17. Function, domain $[2, \infty)$, range $[0, \infty)$

21. Function, domain $[-2, \infty)$, range $[-3, \infty)$

33. y is a function of x

47. y is not a function of x, example for $x = 1$, we have that $y = 1$ or $y = -1$

48. Sasquatches might enter or leave the area, Sasquatches can be born or they can die, range would be $[0, \infty)$

49. Some of them might have multiple emails. Phone numbers would also not be a function since they can also have multiple phone numbers. Social security numbers is unique since it is supposed to be unique.

50. Find the roots of the equation.

1.4 Function Notation

Definition 1.4.1 (Function Notation)**.** If we think of the domain of a function as a set of **inputs** and the range as a set of **outputs**, we can think of a function f as a **process** by which each input x is matches with only one output y. Since the function is completely determined by the input x and thr process f, we symbolize the output with **function notation**: $f(x)$, read 'f of x'. The value of y is completely dependent on the choice of x. For this reason, x is often called the **independent variable**, or **argument** of f, whereas y is often called the **dependent variable**.

1.4.1 Problems

1. $f(x) = \frac{2x+3}{4}$, domain $(-\infty, \infty)$

10. $f(x) = \sqrt{\frac{4}{x}} - 13$, domain $(0, \infty)$

12. (a) $f(3) = 3 - 4(3) = 3 - 12 = -9$
    (b) $f(-1) = 3 - 4(-1) = 3 + 4 = 7$
    (c) $f(\frac{3}{2}) = 3 - 4(\frac{3}{2}) = 3 - 6 = -3$
    (d) $f(4x) = 3 - 4(4x) = 3 - 16x$
    (e) $4f(x) = 4(3 - 4x) = 12 - 16x$
    (f) $f(-x) = 3 + 4x$
    (g) $f(x - 4) = 3 - 4(x - 4) = 3 - 4x + 16 = 19 - 4x$
    (h) $f(x) - 4 = 3 - 4x - 4 = -4x - 1$
    (i) $f(x^2) = 3 - 4x^2$

33.
$$f(0) = \frac{3}{4-0} = \frac{3}{4}$$
$$f(x) \text{ is never equal to } 0$$

37. $(-\infty, \infty)$

63. $A(3) = 9$, $x^2 = 36; x = \pm\sqrt{36} = \pm 6$

1.5 Function Arithmetic

Definition 1.5.1 (Function Arithmetic)**.** Suppose that f and g are functions and x is in both the domain of f and the domain of g.

- The **sum** of f and g, denoted $f+g$, is the function defined by the formula
$$(f + g)(x) = f(x) + g(x)$$
- The **difference** of f and g, denoted $f - g$, is the function defined by the formula
$$(f - g)(x) = f(x) - g(x)$$
- The **product** of f and g, denoted fg, is the function defined by the formula
$$(fg)(x) = f(x)g(x)$$
- The **quotient** of f and g, denoted $\frac{f}{g}$, is the function defined by the formula
$$(\frac{f}{g})(x) = \frac{f(x)}{g(x)}$$

Definition 1.5.2 (Difference Quotient)**.** Given a function f, the **difference quotient** of f is the expression

$$\frac{f(x+h)-f(x)}{h}$$

1.5.1 Problems

1.

$$(f+g)(2) = f(2)+g(2) = 3(2)+1+4-2 = 6+3 = 9$$
$$(f-g)(-1) = f(-1)-g(-1) = 3(-1)+1-4+(-1) = -3+1-4-1 = -7$$
$$(g-f)(1) = g(1)-f(1) = 4-1-3(1)-1 = -1$$
$$(fg)(\frac{1}{2}) = (3(\frac{1}{2})+1)(4-\frac{1}{2}) = (\frac{5}{2})(\frac{7}{2}) = \frac{35}{4}$$
$$(\frac{f}{g})(0) = \frac{1}{4}$$
$$(\frac{g}{f})(-2) = \frac{4-(-2)}{3(-2)+1} = -\frac{6}{5}$$

11.
- Domain $(-\infty,\infty)$, $(f+g)(x) = 3x-1$
- Domain $(-\infty,\infty)$, $(f-g)(x) = x+3$
- Domain $(-\infty,\infty)$, $(fg)(x) = 2x^2-4x+x-2 = 2x^2-3x-2$
- Domain $(-\infty,2)\cup(2,\infty)$, $(\frac{f}{g})(x) = \frac{2x+1}{x-2}$

21.

$$f(x) = 2x-5; \frac{f(x+h)-f(x)}{h} = \frac{2(x+h)-5-2x+5}{h} = \frac{2h}{h} = 2$$

51.

$$(f+g)(-3) = 4-2 = 2$$

1.6 Graphs of Functions

Definition 1.6.1 (The Fundamental Graphing Principle for Functions)**.** The graph of a function f is the set of points which satisfy the equation $y = f(x)$. That is, the point (x,y) is on the graph if and only if $y = f(x)$

Definition 1.6.2 (Zeros of a function)**.** The **Zeros** of a function f are the solutions to the equation $f(x) = 0$. In other words, x is a zero of f if and only if $(x,0)$ is an x-intercept of the graph of $y = f(x)$

Definition 1.6.3 (Testing the Graph of a Function for Symmetry)**.** The graph of a function f is symmetric

- about the y-axis if and only if $f(-x) = f(x)$ for all x in the domain of f
- about the origin if and only if $f(-x) = -f(x)$ for all x in the domain of f

1.6.1 General Function Behavior

Definition 1.6.4 (Increasing, Decreasing and Constant Functions)**.** Suppose f is a function defined on an interval I. We say f is:

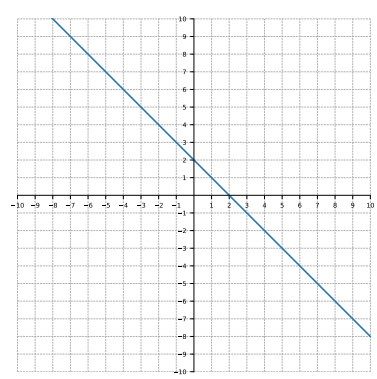
- **increasing** on I if and only if $f(a) < f(b)$ for all real numbers $a, b \in I(a < b)$
- **decreasing** on I if and only if $f(a) > f(b)$ for all real numbers $a, b \in I(a < b)$
- **constant** on I if and only if $f(a) = f(b)$ for all real numbers $a, b \in I$

Definition 1.6.5 (Maximums and Minimums of a Function)**.** Suppose that f is a function with $f(a) = b$

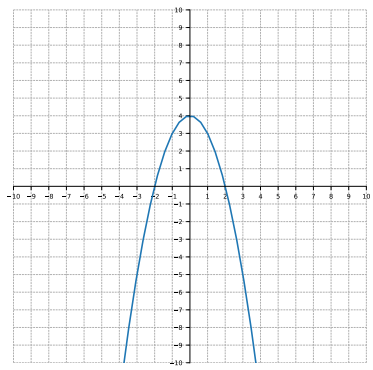
- We say f has a **local maximum** at the point (a, b) if and only if there is an open interval I containing a for which $f(a) \geq f(x)$ for all $x \in I$. The value $f(a) = b$ is called 'a local maximum value of f' in this case.
- We say f has a **local minimum** at the point (a, b) if and only if there is an open interval I containing a for which $f(a) \leq f(x)$ for all $x \in I$. The value $f(a) = b$ is called 'a local minimum value of f' in this case.
- The value b is called the **maximum** of f if $b \geq f(x)$ for all x in the domain of f
- The value b is called the **minimum** of f if $b \leq f(x)$ for all x in the domain of f

1.6.2 Problems

1. Domain $(-\infty, \infty)$, intercepts: $(2, 0)$, Not symmetric



4. Domain $(-\infty, \infty)$, intercepts: $(-2, 0), (2, 0)$, Symmetric about the y-axis



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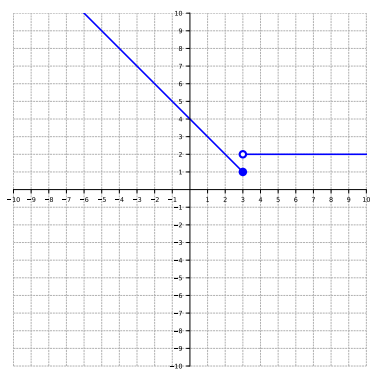


21. Neither even nor odd.

$$f(x) = 7x \neq -7x = f(-x); f(x) = 7x \neq -7x = -f(x)$$

28. Function is even

$$f(x) = 4 - x^2 = f(-x); f(x) = 4 - x^2 \neq= x^2 - 4 = -f(x)$$

1.7 Transformations

Definition 1.7.1 (Vertical Shifts)**.** Suppose f is a function and k is a positive number.

- To graph $y = f(x) + k$, shift the graph of $y = f(x)$ up k units by adding k to the y-coordinates of the points on the graph of f
- To graph $y = f(x) + k$, shift the graph of $y = f(x)$ down k units by subtracting k to the y-coordinates of the points on the graph of f

Definition 1.7.2 (Horizontal Shifts)**.** Suppose f is a function and h is a positive number.

- To graph $y = f(x + h)$, shift the graph of $y = f(x)$ left h units by subtracting h to the x-coordinates of the points on the graph of f
- To graph $y = f(x - h)$, shift the graph of $y = f(x)$ right h units by subtracting h to the x-coordinates of the points on the graph of f

Definition 1.7.3 (Reflections)**.** Suppose f is a function

- To graph $y = -f(x)$, reflect the graph of $y = f(x)$ across the x-axis by multiplying the y-coordinates of the points on the graph of f by -1
- To graph $y = f(-x)$, reflect the graph of $y = f(x)$ across the y-axis by multiplying the x-coordinates of the points on the graph of f by -1

Definition 1.7.4 (Vertical Scalings)**.** Suppose f is a function and $a > 0$. To graph $y = af(x)$, multiply all of the y-coordinates of the points on the graph of f by a. We say the graph of f has been vertically scaled by a factor of a

- If $a > 1$, we say the graph of f has undergone a vertical stretching (expansion, dilation) by a factor of a.
- If $0 < a < 1$, we say the graph of f has undergone a vertical shrinking (compresion, contraction) by a factor of $\frac{1}{a}$

Definition 1.7.5 (Horizontal Scalings)**.** Suppose f is a function and $b > 0$. To graph $y = f(bx)$, divide all of the x-coordinates of the points on the graph of f by b. We say the graph of f has been horizontally scaled by a factor of $\frac{1}{a}$

- If $0 < b < 1$, we say the graph of f has undergone a horizontal stretching (expansion, dilation) by a factor of $\frac{1}{b}$
- If $b > 1$, we say the graph of f has undergone a horizontal shrinking (compresion, contraction) by a factor of b.

Definition 1.7.6 (Transformations)**.** Suppose f is a function. If $A \neq 0$ and $B \neq 0$, then to graph

$$g(x) = Af(Bx + H) + K$$

1. Subtract H from each of the x-coordinates of the points on th graph of f. This results in a horizontal shift to the left if $H > 0$ or right if $H < 0$
2. Divide the x-coordinates of the points on the graph obtained in Step 1 by B. This results in a horizontal scaling, but may also include a reflection about the y-axis if $B < 0$

3. Multiply the y-coordinates of the points on the graph obtained in Step 2 by A. This results in a vertica scaling, but may also include a reflection about the x-axis if $A < 0$

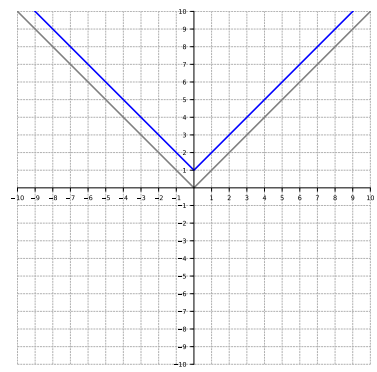
4. Add K to each of the y-coordinates of the points on the graph obtained in Step 3. This results in a vertical shift up if $K > 0$ or down if $K < 0$

1.7.1 Problems

1.

$$(2, 0)$$

19.



64.

$$g(x) = 2f(-x-3) - 1 = 2\sqrt[3]{-x-3} - 1$$

Chapter 2

Linear and Quadratic Functions

2.1 Linear Functions

Definition 2.1.1 (Slope)**.** The **slope** m of the line containing the points $P(x_0, y_0)$ and $Q(x_1, y_1)$ is:

$$m = \frac{y_1 - y_0}{x_1 - x_0} = \frac{\Delta y}{\Delta x}$$

provided $x_1 \neq x_0$.

Definition 2.1.2 (Point-slope form)**.** The **point-slope form** of the line with slope m containing the point (x_0, y_0) is the equation $y - y_0 = m(x - x_0)$.

Definition 2.1.3 (Slope-intercept form)**.** The **slope-intercept form** of the line with slope m and y-intercept $(0, b)$ is the equation $y = mx + b$.

Definition 2.1.4 (Linear function)**.** A **linear function** is a function of the form

$$f(x) = mx + b$$

where m and b are real numbers with $m \neq 0$. The domain of a linear function is $(-\infty, \infty)$.

Definition 2.1.5 (Constant function)**.** A **constant function** is a function of the form

$$f(x) = b$$

, where b is a real number. The domain of a constant function is $(-\infty, \infty)$.

Definition 2.1.6 (Average rate of change)**.** Let f be a function defined on the interval $[a, b]$. The **average rate of change** of f over $[a, b]$ is defined as:

$$\frac{\Delta f}{\Delta x} = \frac{f(b) - f(a)}{b - a}$$

. The line that connects $(a, f(a))$ with $(b, f(b))$ is called the **secant line**.

2.1.1 Exercises

1. $m=3$, $P(3,-1)$; point-slope: $y+1=3(x-3)$, slope-intercept: $y=3x-10$

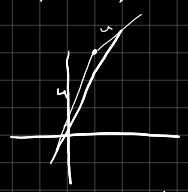
9. $m=-5$, $P(-\sqrt{3}, 2\sqrt{3})$; point-slope: $y-2\sqrt{3}=-5(x-\sqrt{3})$, slope-intercept: $y=-5x+7\sqrt{3}$

11. $P(0,0)$, $Q(-3,5)$; $m=\frac{\Delta y}{\Delta x}=\frac{5-0}{-3-0}=-\frac{5}{3}$ slope-intercept: $y=-5/3\,x$

19. $P(\sqrt{2},-\sqrt{2})$, $Q(-\sqrt{2},\sqrt{2})$; $m=\frac{\sqrt{2}-(-\sqrt{2})}{-\sqrt{2}-\sqrt{2}}=\frac{2\sqrt{2}}{-2\sqrt{2}}=-1$, slope-intercept: $y=-x+2\sqrt{2}$

21. $f(x)=2x-1$, $m=2$, y-intercept: -1, x-intercept: $1/2$

27. $y=2x+1$, $P(-1,3)$;



$$d=\sqrt{(x_1-x_0)^2+(y_1-y_0)^2}=\sqrt{(x_1+1)^2+(y_1-3)^2}$$
$$=\sqrt{(x+1)^2+(2x+1-3)^2}=\sqrt{x^2+2x+1+4x^2-8x+4}$$
$$=\sqrt{5x^2-6x+5}$$
$$16=5x^2-6x+5$$
$$5x^2-6x-11=0,\quad x_1=11/5,\quad x_2=-1,\quad y_1=\frac{27}{5},\quad y_2=-1$$

44. $f(x)=x^3$, $[-1,2]$, $\frac{\Delta y}{\Delta x}=\frac{8-(-1)}{2-(-1)}=3$

50. $f(x)=x^3$, $[x,x+h]$, $\frac{\Delta y}{\Delta x}=\frac{(x+h)^3-x^3}{x+h-x}=\frac{x^3+3x^2h+3xh^2+h^3-x^3}{h}=\frac{3x^2h+3xh^2+h^3}{h}=3x^2+3xh+h^2$

60. $y=-6x+5$, $P(3,2)$; $y_2=-6x+20$

67. $y=\frac{2}{3}x-7$, $P(6,0)$; $y_2=-\frac{3}{2}x+9$